

Practicum No.13**Date: 20/3/2026****Marketing Strategy for a CSE/AIML/DSCS/DS Enterprise****1. Practical Significance**

This practicum equips learners with the ability to conceptualize, design, and implement a marketing strategy tailored to technology-driven startups in Computer Science (CSE), Artificial Intelligence & Machine Learning (AIML), Data Science & Cybersecurity (DSCS), or Data Science (DS) domains. These enterprises operate in fast-evolving competitive ecosystems where product differentiation, technology adoption, and customer trust play crucial roles.

Learners gain the skills to analyze user needs, evaluate competitor technologies, and align product capabilities with market expectations. The practicum promotes strategic thinking and enhances the ability to convert technical insights into compelling value propositions suitable for B2B, B2C, and enterprise markets. Students also learn to create professional marketing components such as problem-solution mapping, customer personas, positioning statements, and go-to-market (GTM) strategies. This includes identifying viable channels such as tech communities, developer forums, SaaS marketplaces, LinkedIn campaigns, GitHub presence, webinars, hackathons, and accelerator platforms.

I. Competency and Industry-Specific Practical Skills

This practicum is expected to develop the following skills/LOs towards the development of the industry-specific competency:

Present the marketing strategy for the chosen CSE, AIML, DSCS & DS enterprise.

- **Develop** a Pitch deck, project proposals for small/medium-scale startups logically and scientifically'. (As stated in the Course Structure)
- a) **Practice** ethics (Affective Domain LO).
- b) **Make** appropriate decisions based on the user insights. (Cognitive Domain LO).
- c) **Prepare** various types of strategy reports and documents required for technical enterprises. (Cognitive Domain LO).

II. Related CO(s) (As stated in the Course Structure)

CO-5 Develop a pitch deck of the identified product/service and its management strategies for ROI.

III. Practical Outcome (PrO) (As stated in the Course Structure)

PrO-13 Present the marketing strategy for **the chosen** CSE, AIML, DSCS & DS enterprise.

IV. Related ADO(s)

- a) **Practice** professional ethics.
- b) **Function** effectively as a team member/leader.
- c) **Practice** integrity and honesty.

V. Minimum Underpinning Theory ((Include relevant content & images for this practicum.))

In fast-growing fields like CSE, AIML, DSCS, and Data Science, marketing strategy is essential to translate technical innovation into market acceptance. Marketing Management Theory, along with STP and the 4Ps, helps position tech products based on performance, scalability, and reliability. These frameworks guide startups in understanding user segments such as developers, enterprises, and educational institutions. Consumer Behavior Theory and Digital Marketing Analytics support data-driven decision-making and channel selection.

Tech enterprises rely on platforms such as GitHub, Kaggle, LinkedIn, and cloud marketplaces to reach target users effectively.

Competitor benchmarking and GTM models help anticipate market shifts and refine messaging. Professionals in companies like Google, Microsoft, Zoho, Freshworks, and Infosys utilize these

models to design impactful strategies. They also use tools such as Miro, Notion, HubSpot, and AI assistants like ChatGPT for campaign planning.

Mastering these theories enables students to craft strong, evidence-based marketing strategies for tech startups. It strengthens analytical, strategic, and communication skills required for building market-ready AI/ML and data-driven solutions.

VI. Practicum Setup

No special lab setup required; group or individual activity in class/computer lab

VII. Resources Required

S.No.	Resource	Broad Specifications	Quantity
1	Computer	With minimum i5 processor, MS Office	1 per student
2	Internet	For market and competitor research	Shared

VIII. Precautions

- a) Maintain confidentiality of proprietary algorithms.
Use professional business language and follow technical documentation standards.

IX. Procedure(Self-Instructional)

1. **Open** an MS Word document titled “Marketing Strategy for the Selected CSE/AIML/DSCS/DS Enterprise.”
2. **Recall** to the selected product/service idea identified during previous ideation, validation, or problem-solving sessions.
3. **List** the key features, technical capabilities, and intended user segments of the chosen tech product/service.
4. **Select** the most suitable target market and customer persona based on user needs, technology readiness, and domain relevance.
5. **Prepare** the foundational marketing framework, including STP (Segmentation, Targeting, Positioning) for structured planning.
6. **Conduct** a detailed marketing strategy analysis by identifying:
 - Segmentation:** User groups such as developers, enterprises, research teams, or students.
 - Targeting:** Primary user segment most likely to adopt your solution.
 - Positioning:** Clear and concise value proposition highlighting differentiation such as accuracy, scalability, automation, or security.
 - 4Ps Strategy:** Product, Price, Place, Promotion elements customized for a technology enterprise.
7. **Carry out** a Channel Strategy Analysis by selecting appropriate digital and B2B channels such as LinkedIn, GitHub, SaaS marketplaces, webinars, hackathons, and email campaigns.
8. **Support** each selected channel and promotional method with evidence collected from industry reports, case studies, analytics sources, and competitor campaigns.
9. **Review** the marketing strategy through peer discussions or instructor consultation, ensuring alignment with the startup’s mission and user expectations.
10. **Refine** the marketing plan using insights from analytics, feedback, and benchmarking — adjusting messaging, channel mix, or positioning as required.
11. **Summarize** the complete marketing strategy, clearly describing the target audience, value proposition, promotional plan, and expected outcomes.
12. **Compile** all strategy components — STP, 4Ps, channel plan, campaign timeline, and visuals — into a structured and professional document.

13. **Save** the finalized MS Word document in the prescribed naming format.
14. **Close** the file and ensure all references, sources, and citations are properly included and formatted.
15. **Submit** the soft copy of the marketing strategy practicum report as per the instructor's submission guidelines.

X. Observations *(To be recorded by students)*

No observations in this practicum. Only results of the tests are there.

XI. Results

The marketing strategy for **Digital Attendance and Productivity Tracker** was developed using structured marketing frameworks such as **STP (Segmentation, Targeting, Positioning)** and the **4Ps marketing model (Product, Price, Place, Promotion)**.

Through **segmentation analysis**, different user groups such as students, teachers, and educational institutions were identified. Among these groups, **schools and colleges were selected as the primary target market**, while teachers and students were considered the main users of the system.

The product is positioned as a **smart digital solution for automated attendance management and productivity tracking**, offering benefits such as improved record accuracy, reduced manual work, and better academic performance monitoring.

Using the **4Ps strategy**, the product features, pricing model, distribution platforms, and promotional activities were clearly defined. Marketing channels such as **educational institution websites, LinkedIn, technology forums, and academic events** were selected to reach the target audience effectively.

XII. Interpretation of Results

The results indicate that a well-planned marketing strategy can significantly improve the **visibility and adoption** of the Digital Attendance and Productivity Tracker.

By identifying the correct target audience and selecting appropriate marketing channels, the startup can effectively reach educational institutions and decision-makers. The use of structured frameworks such as **STP analysis and the 4Ps model** helps define the product's unique value and differentiate it from existing attendance systems.

Digital marketing channels, professional platforms, and educational technology communities can help promote the product and attract early adopters.

XIII. Conclusions

The marketing strategy developed for the **Digital Attendance and Productivity Tracker** provides a clear plan for promoting and introducing the system to educational institutions.

By focusing on the benefits of **automation, efficiency, and productivity monitoring**, the marketing strategy can attract schools, colleges, and training centers that want to modernize their attendance management systems.

Overall, the strategy establishes a strong foundation for launching the product and expanding its adoption in the education technology market.

XIV. Suggested Practical Related Questions *(To be changed for every batch)*

Note: Below are a few questions related to industry-specific higher-order thinking skills (HOT) that teachers must use to ensure students achieve the predetermined course outcomes

1. Evaluate the extent to which the company's current market positioning strengthens its credibility among enterprise clients adopting AI/ML or cybersecurity solutions.
2. Analyze the effectiveness of the company's differentiation strategies compared to industry leaders

such as Google AI, AWS, and GitHub in terms of messaging, feature clarity, and community engagement.

3. Assess the influence of the company's developer-focused outreach (e.g., GitHub repositories, APIs, SDKs) on adoption and long-term brand loyalty.
4. Justify whether the company's pricing model (subscription, pay-as-you-go, enterprise licensing) aligns with global competitor practices and customer willingness to pay.
5. Predict future marketing challenges the company might encounter due to rapid advancements in automation, data privacy regulations, or generative AI trends.
6. Examine the impact of the company's participation in hackathons, tech expos, and accelerator programs on overall brand visibility and lead generation.
7. Design a refined marketing funnel that increases the company's conversion rate from awareness to enterprise purchase, using digital analytics and customer behavior insights.
8. Recommend improvements in the company's content marketing—whitepapers, case studies, webinars, benchmarks—to enhance technical credibility and customer trust.
9. Propose strategic collaborations with cloud providers, academic institutions, or AI research labs to expand the company's market reach and innovation capability.
10. Formulate a data-driven roadmap that incorporates user feedback, competitor intelligence, and performance metrics to strengthen the company's overall marketing strategy.

XV. References/Suggestions for Further Reading

Gassmann, O., Frankenberger, K., & Csik, M. (2021). *The Business Model Navigator: 55 Models That Will Revolutionize Your Business*. Pearson.

XVI. Suggested Assessment Scheme (*The performance indicators and their weightages will differ from practicum-to-practicum*)

The performance indicators given serve as a guideline for assessing the 'process-related skills/LOs' (marks to be awarded in real-time in the laboratory by the faculty member, as these cannot be measured after the practicum is over) and 'product-related skills/LOs'

Note: The weightage for the process-related skills and product-related skills will change depending on the PrO, COs, and the competency related to the concerned course.

Performance indicators		Maximum Weightage	Marks Obtained
Process related skills:- 40%			
1	<u>Use</u> structured method to identify factors in SWOC.	10	
2	<u>Function</u> effectively as a team member.	10	9 (say)
3	<u>Practice</u> ethical decision-making and critical thinking.	10	
4	<u>Participate</u> in discussion and idea refinement actively.	10	
Product related skills: - 60%			
5	<u>Complete</u> the SWOT/C analysis with logical depth and clarity	20	
6	<u>Identify</u> realistic and significant constraints affecting feasibility	15	
7	<u>Refine</u> the idea based on analytical insights and constraints	15	
8	<u>Present</u> the final report with clarity, structure, and on-time	10	
Total		100	

To be filled by Student:

S.No.	Name of the Student	Register No.
1	Neavis Rishva. J	URK25CS1236

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Date of Evaluation
Process-Related Skills	Product-Related Skills	Total		

